

MA 548: Abstract Algebra

Course Code	MA 548
Course Name	Abstract Algebra
L-T-P-C	3-1-0-4
Intended for	M.Sc./M.S./PhD/B.Tech
Distribution	Discipline Elective for M.Sc. Applied Mathematics and Elective for Other Disciplines
Mutual Exclusion	MA 251
Approval	59th BoA (Apr 2025)

Course Contents

- **Unit I:** Binary operation, and its properties, Definition of a group, Examples and basic properties, Subgroups, Cyclic groups, Order of a group. [6 Lectures]
- **Unit II:** Permutation groups, Isomorphism and its properties, Cayley's theorems, Coset of a subgroup, Lagrange's theorem. [6 Lectures]
- **Unit III:** Normal subgroups, Quotient group. Homomorphisms, Kernel Image of a homomorphism, Isomorphism theorems, Direct product of groups, Structure of finite abelian groups. [6 Lectures]
- **Unit IV:** Group action and class equation, Sylow's Theorems and its applications, Solvable and Nilpotent groups, Lower and Upper central series. [6 Lectures]
- **Unit V:** Rings: definition, examples and basic properties, Zero divisors, Integral domains, Fields, Characteristic of a ring, Quotient field of an integral domain, Subrings, Ideals, Quotient rings. [6 Lectures]
- **Unit VI:** Isomorphism theorems, Prime ideal, Maximal ideals, Ring of polynomials, Division Algorithm and its Consequences, Irreducibility test, Prime elements, Irreducible elements and their properties, UFD, PID and Euclidean domains. [8 Lectures]
- **Unit VII:** The Fundamental Theorem of Field Theory, Splitting Fields, Zeros of Irreducible Polynomials. [4 Lectures]

Laboratory Modules

Weekly tutorials are required to practice the questions based on the contents discussed in the class

Text Books

1. Joseph Gallian, **Contemporary Abstract Algebra**, 7th Edition.
2. N. Herstein, **Topics in Algebra**, Wiley Eastern Ltd., 1975.

References

1. M. Artin, **Algebra**, Prentice Hall of India, 1994.
2. J. B. Fraleigh, **A first Course in Abstract Algebra**, Narosa Publishing House, 2003.
3. Klima, Sigmon and Stitzinger, **Applications of Abstract Algebra with Maple and MATLAB**, Second Edition, 2006.